



Capital University of Science and Technology

Department of Software Engineering

SE4563– Cloud Computing

Mid Term Exam (BS SE)

Semester: Spring 2025

Date: 28/04/2025

Max Marks: 20

Time: 90 Minutes

Instructors: Mr. Zaheer Ahmad

Instructions:

- This exam contributes 20% weightage to your final evaluation.
- Attempt all questions and answer briefly on the answer sheet.

Name:

Reg. No.

Part I

(CLO-1) (10 Marks)

Question No. 1

(4+3+3=10)

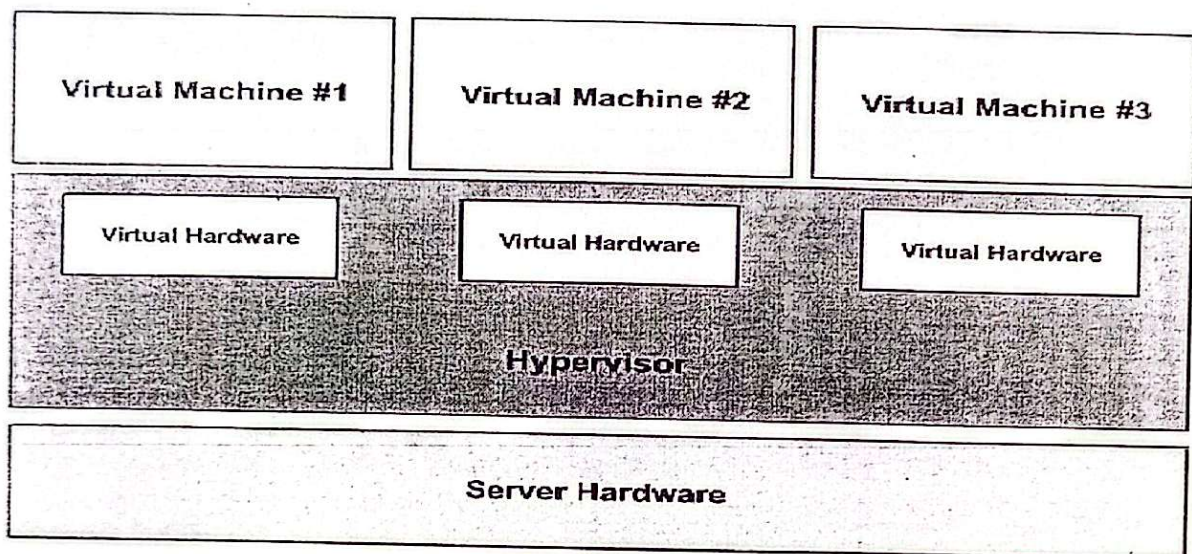
- a) For each of the following situations, **identify** the cloud deployment model (Public, Private, or Hybrid) that represents the **best fit** for their specific needs and then **explain why** that model is the most suitable choice.
1. A government agency requiring strict control over sensitive data.
 2. A startup needing scalable and cost-effective infrastructure for its web application.
 3. A large enterprise using its own data center for critical applications while leveraging external resources for peak loads.
- b) A team of student developers at Capital University of Science and Technology is working on a web application for their final year project. They need a platform to build, test, and deploy their application. They require support for their chosen programming language (Python), a database service, and easy deployment capabilities without having to configure servers or manage operating systems." **Identify** the most appropriate cloud service model for their needs and **explain why**, focusing on the advantages it offers for application development and deployment.
- c) A company wants to move its existing email service to the cloud. **Identify** the most suitable cloud service model for this purpose and **explain why** it is the best fit.

Part II
(CLO-2) (10 Marks)

Question No. 2

(5+5=10)

- a) The diagram below illustrates a common architecture for hardware-based virtualization. Refer to this diagram to answer the following questions:
1. Describe how the guest operating systems within the VMs interact with the underlying physical hardware, based on the diagram.
 2. Explain the role of the "Virtual Hardware" layer as presented in the diagram. What is its purpose in the context of the virtual machines?
 3. Briefly describe how the architecture depicted in the diagram allows multiple independent operating systems to run concurrently on a single "Server Hardware."



- b) A software company in Rawalpindi has developed a web application with several independent parts: a user interface, an ordering system, and a payment procedure. They need to deploy these parts onto their servers. They are considering two main ways Virtual Machines (VMs) and Containers (using Docker) to isolate and run these different parts:
1. If the company chooses to use containers with Docker, what is one key advantage they would gain in terms of how quickly they can start and stop each part of their application compared to using separate Virtual Machines? **Explain** why.
 2. As their application becomes more popular, they might need to run multiple copies of each part (e.g., several copies of the ordering system to handle many orders at once). What technology is commonly used to automate the management and scaling of these containers across multiple servers? (Just name the technology).
 3. Briefly describe one situation where the company might still choose to use a Virtual Machine for one of their application parts instead of a container.